

Resources allowed: Calculator and 1 x A4 page one side
Time Allowed: 45 mins

Score: /35 %

1. Find the five number summary for the following set of data. [5]

6 10 10 12 16 21 25 26 27 28 35

Minimum	Q_1	Median	Q_3	Maximum
6 ✓	10 ✓	21 ✓	27 ✓	35 ✓

2. Find the five number summary for the following set of data: [5]

40 62 63 64 64 67 72 75 78 80 81 82 85 87 90 93

Q_1 Q_3

minimum: 40 ✓

$Q_1: 64 \quad \checkmark$

median: 76.5 ✓

Q3: 83-5 ✓

Maximum: 93 ✓

3. Draw the boxplot for the following set of data.

[8]

Mathematics scores for a Year 10 end of Year Examination (%)

Scores	42	47	52	55	55	58	61	62	64	65	67	68	72	80	88	92
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55 63 70

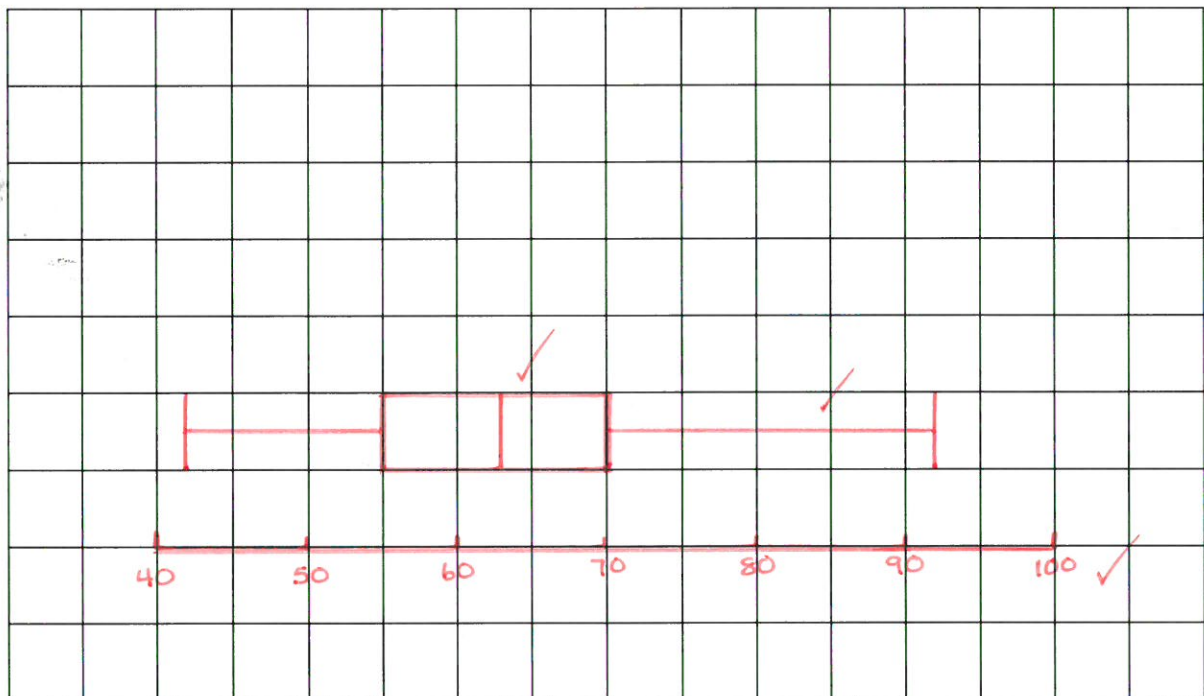
min: 42 ✓

Q₁: 55 ✓

median: 63 ✓

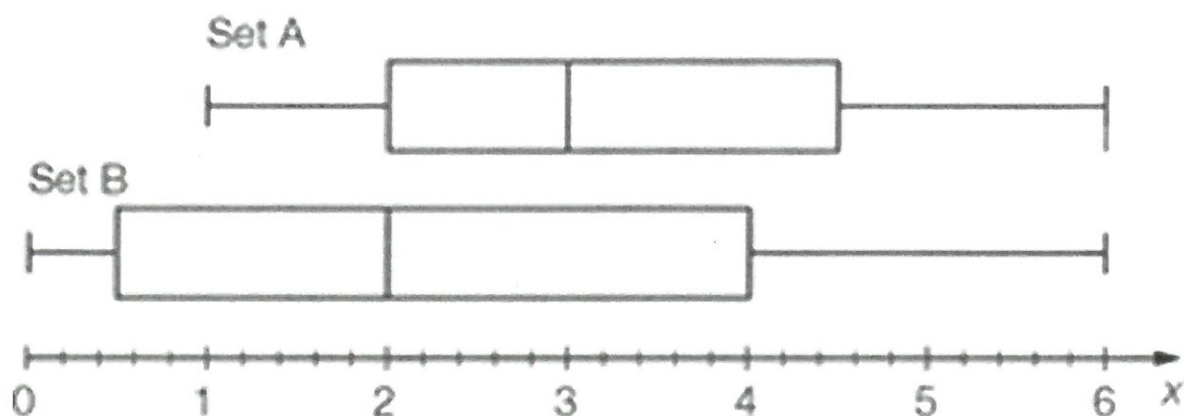
Q₃: 70 ✓

max: 92 ✓



4. The following boxplots are mathematics quiz scores, out of 8, for two classes, A and B.

[8, 4 = 12]



(a) For each class find the five number summary and the interquartile range (IQR)

	Set A	Set B
Minimum	1 ✓	0 ✓
Q ₁	2 ✓	$\frac{1}{2}$ ✓
Median	3 ✓	2 ✓
Q ₃	$4\frac{1}{2}$ ✓	4 ✓
Maximum	6 ✓	6 ✓
IQR	$4\frac{1}{2} - 2 = 2\frac{1}{2}$ ✓✓✓	$4 - \frac{1}{2} = 3\frac{1}{2}$ ✓✓✓

($\frac{1}{2}$ each tick)

(b) Compare the statistics for the two classes. Which did the best? Justify your answer.

Minimum 4 valid comments.

- Set A did best.
- Set A has a smaller range.
- The middle 50% of students achieved higher than the median of set B.
- 75% of set A achieved more than 2/8, only 50% of set B achieved this score.
- Any other reasonable response.

5. Write a set of data with at least 10 values that has a lowest value of 6, a highest value of 20, lower quartile of 9, upper quartile of 15 and median of 12. [5]

Various responses are possible.

$\boxed{6}, 7, 8, \boxed{9}, 10, 12, \boxed{12}, 12, 14, \boxed{15}, 17, 19, \boxed{20}$

END OF TEST